

Smart Lender: Project Requirements Specification

1. Functional Requirements (FR)

The core classification system must fulfill the following operational processing behaviors:

- **FR-1 (Profile Ingestion):** The system must securely accept multi-parameter categorical arrays representing applicant socio-demographic contexts (gender, married state, dependency indices, education, and geographic location tags).
- **FR-2 (Financial Metrics Compilation):** The analytical backend engine must parse continuous input channels representing primary base revenue stream values, coapplicant financial indicators, overall requested application sums, and term limits.
- **FR-3 (Dynamic Engine Switching):** The interface layer must support hot-swapping configuration flags across four independent pre-trained evaluation frameworks (Decision Tree, Random Forest, KNN, XGBoost) mid-session.
- **FR-4 (Relational Storage Sink):** Every unique evaluation lifecycle run must generate structured execution payloads matching transactional data entities (prediction identity numbers, continuous probabilities, timestamps).

2. Non-Functional Requirements (NFR)

System deployment configurations must comply with institutional runtime policies:

- **NFR-1 (Portability Standard):** Microservices must run encapsulated inside an isolated, immutable Docker runtime container, making them completely host-independent.
- **NFR-2 (Performance Optimization):** Average inference runtime loop processing delays must remain under 100 milliseconds during high-frequency parallel transaction modeling.
- **NFR-3 (UI Fluidity):** Client rendering templates must dynamically adjust presentation margins to provide clear, user-friendly forms across different mobile or monitor viewports.

3. System Stakeholder Matrix

User Category	Primary Platform Touchpoint	Expected Operational Value Target
Lending Officer	Web Entry Input Forms Form View	Instant credit risk assessment during live validation sessions.
Risk Analyst	Multi-Model Comparative Benchmarks	Evaluating data distribution correlations and setting policy boundary terms.

User Category	Primary Platform Touchpoint	Expected Operational Value Target
Data Scientist	Core Script Pipelines & Docker Assets	Validating accuracy thresholds, scaling variables, and mitigating class skew.